Duhamel Neural Operator for Learning Phase-Field Dynamics

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Abstract

Traditional ferroelectric phase-field simulators are used to predict incubation, nucleation, and saturation trajectories, a task that has been challenging due to their high computational cost. Although machine learning surrogates have recently become increasingly adopted due to their quick inference times, they struggle to approximate the amplification of incubation noise. Motivated by Duhamel's principle, we introduce the Duhamel Neural Operator (Duhamel-NO), a deep learning surrogate that combines a Fourier state-space model (SSM) and a next-step U-Shaped Neural Operator (UNO) to simulate nucleation behavior. By conditioning Duhamel-NO on the Landau coefficients α, β, γ, and initial noise fields, we reproduce the reference solver's (FERROX) trajectories up to three orders of magnitude faster. Importantly, Duhamel-NO exhibits the most physically consistent predictions and significantly lower error than the best of seven tuned baselines, achieving a near-perfect pattern zero-shot generalization on unseen materials, initializations, and free energy landscapes. Code is available at https://github.com/matthew-ju/Duhamel-Neural-Operator.

1 Introduction

Ferroelectrics store information in the direction of their spontaneous polarization, and an external electric field can switch this polarization, which then retains its state after the field is removed. This unique behavior is set by how polarization domains incubate, nucleate, and saturate. Hafnia-based ferroelectric films stay ferroelectric at thicknesses compatible with silicon processing. As such, they are increasingly being adopted as alternatives for memory and logic. The standard model for their domain dynamics is the phase-field method: the out-of-plane polarization $P_z(\mathbf{r}, t)$ relaxes by time-dependent Ginzburg–Landau (TDGL) gradient flow, coupled to the electrostatics of the device stack [Chen, 2002]. Graphics processing unit (GPU) solvers such as FERROX [Kumar et al., 2023] simulate phase-fields at high accuracy, but one 3D trajectory is computationally expensive. This is consequential, as a design that studies materials with a grid of n values of each Landau coefficient needs n^3 trajectories, which adds up to hundreds, if not thousands, of hours.

A trajectory has three stages and two regimes. The linear regime is entirely in the incubation stage, while the nonlinear regime consists of the nucleation and saturation stages. The trajectory starts

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from small random noise, about 0.5% of the saturated polarization P_{S}. During incubation, a band of Fourier modes of this noise grows exponentially while the other modes decay. Domains then nucleate, and finally they coarsen slowly. The final domain pattern is determined by which noise modes win during incubation. Thus, a standard autoregressive model that performs poorly in the first few frames cannot recover the right pattern. This is where the state-of-the-art neural surrogates are weakest, as they have to use data to learn a response that grows on some modes and decays quickly on others, on a field that shrinks to three orders of magnitude below the saturated one.

Our approach starts from the equation. The right-hand side of the TDGL equation splits exactly into a linear part and a nonlinear part. Duhamel's principle then writes the solution one step later as a linear propagator applied to the current state, plus an integral of the nonlinear force. The linear propagator is exactly a diagonal Fourier state-space model (SSM), and we compute it in closed form. A neural operator learns only the integral term. This term plays the same role as the higher-order Landau terms in the free energy: it is a correcting force that stops the exponential growth and saturates the field. The result, the Duhamel Neural Operator (Duhamel-NO), is a single neural operator whose architecture contains the structure of the TDGL solution.

The same structure makes conditioning on the material natural. The Landau coefficient α enters only the linear part, and the coefficients β and γ enter only the nonlinear part. We train one model on a Latin-hypercube set of materials and test it without retraining on held-out materials, materials outside the training range, and wells of a different shape. Every test trajectory starts from unseen noise.

Contributions.

1. **Duhamel-NO, a neural operator structured by Duhamel's principle** (Section 4). Its linear part is a diagonal Fourier SSM fitted by regression; a network learns the nonlinear correction.
2. **Zero-shot generalization across materials and noise** (Section 5.1). On 256 held-out materials it reaches pattern correlation 0.982 at frame 400. On 300 materials outside the training range it reaches 0.978. Its error is 23–26$\%$ lower than that of the best of seven tuned baselines at frame 400, and $2.61\times$ lower at frame 1.
3. **Physical fidelity and speed** (Sections 5.2 and 5.3). The predicted fields give better nucleation times, saturation polarization, electrostatic fields and free energy than every baseline, at about $1814\times$ the throughput of the solver.

2 Related Work

Exponential integrators and neural operators. Integrating the stiff linear part of an equation exactly, and approximating only the nonlinear term, is the idea behind exponential time differencing [Cox and Matthews, 2002, Kassam and Trefethen, 2005, Hochbruck and Ostermann, 2010]. Our update is the exponential-Euler member of this family. Several neural operators also integrate a Fourier symbol exactly: neural stochastic partial differential equations (SPDEs) use the mild (Duhamel) form of the equation [Salvi et al., 2022], and SSNO and SGNO (stable spectral and spectral generator neural operators) integrate a learned, dissipative symbol [Zhang et al., 2025, Li et al., 2026]. Duhamel-NO differs in three ways. Its linear operator must *grow* on a band of modes, because this instability seeds the pattern. The operator is *measured* from simulations rather than learned, and it couples the depth layers of a 3D film. And it is interpolated across materials.

State-space models and data-driven linear operators. Deep state-space models became practical once the state transition was made diagonal [Gu et al., 2022a,b, Gupta et al., 2022, Orvieto et al., 2023]. The TDGL Fourier SSM is exactly such a model (Proposition 1). It can therefore be computed in closed form from snapshot pairs, the exact dynamic mode decomposition (DMD) or Koopman construction [Schmid, 2010, Tu et al., 2014, Kutz et al., 2016]. Linearizing about the uniform state is the classical theory of spinodal decomposition [Cahn and Hilliard, 1958, Cahn, 1961, Allen and Cahn, 1979].

Ferroelectric and phase-field surrogates. Earlier ferroelectric surrogates roll the trajectory forward with one learned model, such as an autoregressive network [Alhada-Lahbabi et al., 2024] or a long short-term memory (LSTM) network over principal components [Alhada-Lahbabi et al., 2023]. Surrogates for other phase-field systems also fit one model across the whole horizon [Bonneville

et al., 2025a,b, Oommen et al., 2022], and some condition on material parameters [Fantasia et al., 2026, Hamdi et al., 2025]. To our knowledge, earlier ferroelectric surrogates train and test at fixed coefficients, and on initial noise seen during training. Such leakage inflates accuracy [Kapoor and Narayanan, 2023, McGreivy and Hakim, 2024], so every test trajectory here has its own noise.

3 Problem Setup

TDGL phase-field dynamics. The polarization relaxes by gradient flow of a Ginzburg–Landau free energy:

$$\frac{\partial P_z}{\partial t} = -\Gamma \frac{\delta F}{\delta P_z}, \qquad F[P_z] = \int \Big[\underbrace{\tfrac{\alpha}{2} P_z^2 + \tfrac{G_{11}}{2} \|\nabla P_z\|^2}_{\text{quadratic}} + \underbrace{\tfrac{\beta}{4} P_z^4 + \tfrac{\gamma}{6} P_z^6}_{\text{higher order}} \Big] d\mathbf{r} + U_{\text{elec}}[P_z]. \tag{1}$$

Here U_{elec} is the electrostatic energy, which is quadratic in P_z and is computed from a Poisson equation for the whole stack. The coefficient $\alpha < 0$ sets the depth of the double well, and β and γ set its shape; $\beta < 0$ gives a first-order well, which only the sextic term keeps stable.

Device and simulations. We simulate a metal–ferroelectric–insulator–metal stack with FERROX: a 5 nm ferroelectric film (ε_{fe}=24) on a 4 nm dielectric (ε_{de}=10) between shorted electrodes, periodic in-plane, with G_{11}=10^{-9} V m^3/C and Γ=100 $(\Omega\,\text{m})^{-1}$. The state is P_z on a $64 \times 64 \times 10$ grid. Each run starts from small uniform random noise and is saved for 401 frames, with a frame interval per material chosen so that the frames cover about $1.25\times$ its time to saturation.

Datasets. **D1** contains 1001 materials, drawn as a Latin hypercube in the well depth $|\alpha| \in [1.6, 4.0] \times 10^9$ V m/C, the saturated polarization $P_{\text{S}} \in [0.10, 0.43]$ C/m^2, and the well shape $r \in [0, 1.4]$, the share of the restoring force that comes from the sextic term ($r > 1$ means $\beta < 0$); β and γ are solved from these coordinates. 745 materials are used for training (682 train, 63 validation) and 256 are held out: 82 inside the sampled box and 174 at its edges, including every well with $r \geq 1.15$. **D2** contains 300 further materials with independently drawn α, β, γ in five regions: small P_{S}; large P_{S} (up to $2.2\times$ the training maximum); large $|\alpha|$ with small P_{S}; first-order wells with r up to 2; and all of these together. D2 is tested without retraining. Each material is simulated once.

Unseen noise. FERROX generates the same initial field for the same seed, whatever the material, so a model can memorize shared seeds: with eight shared seeds, a model scores 0.99 on a seen seed but only 0.66–0.82 on an unseen one. We therefore give *every trajectory its own seed*. The largest correlation between a test and a training initial field is 0.023.

Allen–Cahn. To test transfer to new grids and wall widths, we also use the 2D Allen–Cahn equation $\partial_t u = -M(au + bu^3 - e\nabla^2 u)$ with $a < 0$, which shares the noise-seeded structure of TDGL. We generated these data ourselves with an exponential-integrator solver, which favours an exponential-Euler model such as Duhamel-NO, on a periodic 64^2 grid from small random noise. The grid spacing in wall widths, $\Pi_h = \Delta x/\mu$ with $\mu = \sqrt{e/|a|}$, lies in $[0.30, 0.70]$ for training and in $[0.85, 1.20]$ for the *new physics* split. The in-range test materials are also simulated on 32^2 and 128^2 grids (Section G).

Metrics. We report the *pattern correlation* of a single frame, $\text{corr}(A, B) = \sum_i (A_i - \bar{A})(B_i - \bar{B}) / [\sum_i (A_i - \bar{A})^2 \sum_i (B_i - \bar{B})^2]^{1/2}$, and the relative error $\|A - B\|_2 / \|B\|_2$, where t counts saved frames and t=400 is the settled state. Because a low field error does not guarantee correct physics, we also recompute from each prediction the quantities a materials scientist would extract: nucleation time, saturation polarization, domain size, wall density, the distribution of P/P_{S}, the electrostatic potential and depolarization field, and the free energy. We fixed this list before evaluating any model. Every prediction is rolled out from the t=0 field, with no solver frame.

4 Method

4.1 Duhamel's principle for TDGL

Taking the variational derivative in Equation (1) splits the right-hand side into two parts. The quadratic energy gives a term that is linear in P_z. The higher-order energy gives a nonlinear term:

$$\partial_t P_z = \mathcal{L}P_z + \mathcal{N}(P_z), \quad \mathcal{L}P_z = -\Gamma\big(\alpha P_z - G_{11}\nabla^2 P_z - E_z[P_z]\big), \quad \mathcal{N}(P_z) = -\Gamma\big(\beta P_z^3 + \gamma P_z^5\big). \tag{2}$$

E_z is the depolarization field, which is linear in P_z. This split is exact.

The original principle. Duhamel's principle (variation of constants) solves a *linear* equation with a *given* source $f(t)$. For $\partial_t u = \mathcal{L}u + f(t)$, the state one frame of length τ later is

$$u(t+\tau) = e^{\tau\mathcal{L}}u(t) + \int_0^\tau e^{(\tau-s)\mathcal{L}} f(t+s)\, ds. \tag{3}$$

It follows by multiplying the equation by $e^{-s\mathcal{L}}$, which gives $\frac{d}{ds}\big(e^{-s\mathcal{L}}u\big) = e^{-s\mathcal{L}}f$, and integrating over the frame. The first term is the response to the current state, and the second is the accumulated response to the source.

Applying it to TDGL. In Equation (2), the nonlinear term plays the role of the source, $f(s) = \mathcal{N}(P_z(t+s))$. The difference from the textbook case is that this source is not given: it depends on the unknown solution. Equation (3) then becomes

$$P_z(t+\tau) \;=\; \underbrace{e^{\tau\mathcal{L}}\, P_z(t)}_{\text{Fourier SSM}} \;+\; \underbrace{\int_0^\tau e^{(\tau-s)\mathcal{L}}\, \mathcal{N}\big(P_z(t+s)\big)\, ds}_{\text{nonlinear correction}}. \tag{4}$$

This is still exact, but implicit: the integral depends on the path $P_z(t+s)$ inside the frame, which is known only after the equation is solved, and one frame spans many solver steps.

The two terms have different physical roles. $\mathcal{L}$ is unstable on a band of wavenumbers, so the first term amplifies the noise modes that seed the domain pattern. The force $\mathcal{N}$ comes from the higher-order Landau terms and is of order P_z^3. It is negligible while the field is small, and once the field grows it opposes further growth and saturates the field at P_S (for $\beta < 0$ the quartic term pushes the other way, and the sextic term provides the restoring force). The integral is therefore the correction that turns exponential growth into domains, in the same way that the higher-order terms of F bound the energy from below.

4.2 Duhamel-NO

Equation (4) is exact but implicit. Because the dynamics are deterministic and do not depend explicitly on time, the state one frame later is a function of the current state, $P_z(t+\tau) = \Phi_\tau(P_z(t))$, and so is the integral. Substituting $u = \tau - s$ gives $\int_0^\tau e^{(\tau-s)\mathcal{L}} ds = \tau\varphi_1(\tau\mathcal{L})$, with $\varphi_1(z) = (e^z - 1)/z > 0$ for real z; since $\mathcal{L}$ is symmetric (Section 4.3), this operator is invertible. The integral can therefore be written exactly as $\tau\varphi_1(\tau\mathcal{L})$ applied to an *effective force*, the constant force that gives the same frame as the true, changing one:

$$P_z(t+\tau) = e^{\tau\mathcal{L}}P_z(t) + \tau\varphi_1(\tau\mathcal{L})\, N_{\text{eff}}(P_z(t)), \qquad N_{\text{eff}} = \big[\tau\varphi_1(\tau\mathcal{L})\big]^{-1}\big(\Phi_\tau(P_z) - e^{\tau\mathcal{L}}P_z\big). \tag{5}$$

If $\mathcal{N}$ were constant over the frame, $N_{\text{eff}} = \mathcal{N}(P_z(t))$ and Equation (5) would be the exponential-Euler scheme [Cox and Matthews, 2002, Hochbruck and Ostermann, 2010]; in general $N_{\text{eff}} = \mathcal{N}(P_z(t)) + O(\tau)$ on modes that are not stiff (Section A.2), and during incubation $N_{\text{eff}} \approx 0$. N_{eff} has no closed form, but it depends only on the current state and the material, so we learn it with a neural operator $N_\psi(v; \mathbf{m})$, where v is the field in Fourier space and $\mathbf{m}$ are the material numbers (Figure 1). With L the Fourier symbol of $\mathcal{L}$ and θ the frame interval, this gives the model

$$v_{n+1} \;=\; e^{\theta L}\, v_n \;+\; \theta\, \varphi_1(\theta L)\, N_\psi(v_n; \mathbf{m}). \tag{6}$$

The first term moves every Fourier mode by its exact linear factor. The filter φ_1 damps the network's output on stiff modes, as the true integral does. The network's last layer starts at zero, so the untrained model is the exact linear solution. The only approximation is $N_\psi \approx N_{\text{eff}}$; any error in the measured L (Section 4.3) is passed on to the network.

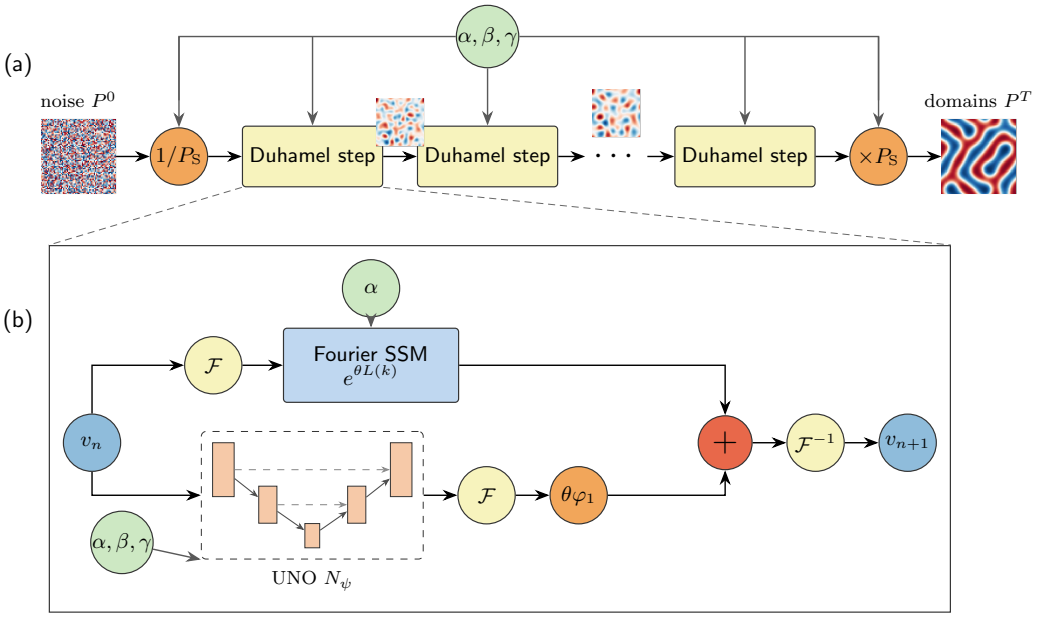


Figure 1: **Duhamel-NO.** (a) The initial noise P^0 is put in material units and advanced by repeated Duhamel steps; the material (α, β, γ) conditions every step. (b) One step (Equation (6)): the Fourier SSM $e^{\theta L(k)}$, set by α, multiplies every Fourier mode of v_n by its exact linear factor. The U-shaped UNO N_ψ (resolution $64^2 \to 32^2 \to 16^2$ and back, with skip connections), conditioned on material numbers computed from (α, β, γ), gives the nonlinear correction, which is filtered by $\theta\varphi_1(\theta L)$ and added in Fourier space.

4.3 Fourier SSM

$\mathcal{L}$ is translation-invariant in the plane of the film, but the electrostatics couple the ten layers. The in-plane Fourier modes therefore evolve independently: for each wavevector k, the ten layer amplitudes $\hat{P}_z(k,t) \in \mathbb{C}^{10}$ obey

$$\partial_t \hat{P}_z(k,t) = L(k)\,\hat{P}_z(k,t), \qquad L(k) = -\Gamma\big[(\alpha + G_{11}|k|^2)\,I + C(k)\big], \tag{7}$$

where $C(k)$ is the 10×10 layer-coupling matrix (finite differences across the film and the depolarization field). $L(k)$ is symmetric, and a mode grows when one of its eigenvalues is positive: $\alpha < 0$ drives growth, the gradient term damps large $|k|$ and the depolarization field damps small $|k|$, so growth is confined to a band that sets the domain size. Because $L(k)$ is real symmetric up to the factor $-\Gamma$, it has real eigenvalues $\lambda_i(k)$ and orthonormal eigenvectors $V(k)$, and integrating Equation (7) over one frame gives $B(k) = e^{\theta L(k)} = V(k)\,\mathrm{diag}\big(e^{\theta\lambda_i(k)}\big)V(k)^\top$. Since α multiplies the identity, $L(k;\alpha) = L(k;\alpha') - \Gamma(\alpha - \alpha')I$, and β, γ do not appear in $L(k)$.

Proposition 1 (The linear propagator is a diagonal Fourier SSM)**.** Write $B(k) = V(k)\Lambda(k)V(k)^{-1}$. Then the linear dynamics are exactly the diagonal, linear, time-invariant state-space recurrence

$$h^0(k) = V(k)^{-1}\hat{P}_z{}^0(k), \qquad h^{t+1}(k) = \Lambda(k)\,h^t(k), \qquad \hat{P}_z{}^t(k) = V(k)\,h^t(k),$$

with $A = \Lambda$, $C = V$ and $D = 0$, and with the initial noise as an impulse input. This is the form of the diagonal SSMs S4D (diagonal structured state space), DSS (diagonal state spaces) and LRU (linear recurrent unit) [Gu et al., 2022b, Gupta et al., 2022, Orvieto et al., 2023].

The proof is in Section A.3. Each in-plane mode is an independent copy of this system, so one Fourier SSM step costs one 10×10 matrix–vector product per mode.

Measuring $B(k)$. We measure $B(k)$ rather than assemble $C(k)$, because the solver's discrete operators differ from the continuum ones. While the field is small, consecutive frames obey $\hat{P}_z{}^{t+1} = B\hat{P}_z{}^t$. For each of 46 radial bins of $|k|$, we stack the frame pairs $t = 0, \ldots, 7$ of all modes in the bin

Algorithm 1 Duhamel-NO: rollout from initial noise

Require: initial noise P^0; coefficients (α, β, γ); number of frames T; trained N_ψ

$P_{\mathrm{S}}, \lambda, r \leftarrow$ closed-form material units (Section C.3); $\theta \leftarrow \Delta t_{\mathrm{frame}}/\lambda$

$\mathbf{m} \leftarrow (\log\theta, \log P_{\mathrm{S}}, r, \log\Pi_h)$ ▷ $\Pi_h = \Delta x/\mu$: grid spacing in wall widths

for every in-plane mode k: $E(k) \leftarrow e^{\theta L(k)}$, $\Phi(k) \leftarrow \theta\varphi_1(\theta L(k))$ ▷ $L(k)$ from Algorithm 3

$v \leftarrow P^0/P_{\mathrm{S}}$

for $n = 0, \ldots, T-1$ **do**

  $s \leftarrow \mathrm{rms}(v)$; $h \leftarrow s \cdot N_\psi(v/s,\ \log_{10} s,\ \mathbf{m})$ ▷ nonlinear correction

  $\hat{v} \leftarrow \mathrm{FFT}(v)$; $\hat{h} \leftarrow \mathrm{FFT}(h)$ ▷ in-plane transform of the ten layers

  $\hat{v}(k) \leftarrow E(k)\,\hat{v}(k) + \Phi(k)\,\hat{h}(k)$ for every mode k ▷ Equation (6)

  $v \leftarrow$ bounded $\mathrm{IFFT}(\hat{v})$; $P^{n+1} \leftarrow P_{\mathrm{S}}\, v$

return $P^1, \ldots, P^T$

into matrices X and Y, and Equation (8) gives B_b in closed form (exact DMD [Schmid, 2010, Tu et al., 2014], small ridge ϵ):

$$B_b = \arg\min_B \|Y - BX\|_F^2 = YX^{\mathsf{H}}\big(XX^{\mathsf{H}} + \epsilon I\big)^{-1}. \tag{8}$$

For a new material we average the generators $L_b = \theta^{-1}\log B_b$ of the eight training materials closest in $\log|\alpha|$. The Fourier SSM alone reproduces frame 1 at correlation 0.997 on validation materials (Algorithms 2 and 3 in Section B).

4.4 The Neural Operator

Material units. The network works in each material's own units: P_z in units of P_{S}, and time in units of the linear growth time λ. Both come from closed-form laws in (α, β, γ) that account for depolarization (Section C.3). In these units, the nonlinear force of every material has the same form, $-(1-r)v^3 - rv^5$. So β and γ enter the correction through a single shape number r. This change of units alone, before any training, reduces the spread (coefficient of variation) of the nucleation time across D1 materials from 0.74 to 0.054.

The network. N_ψ is a U-shaped neural operator (UNO) [Rahman et al., 2023]: a Fourier neural operator arranged like a U-Net. Each block multiplies the lowest Fourier modes of its input by learned weights, adds a pointwise linear map, applies a Gaussian error linear unit (GELU), and resamples the field to a coarser or finer grid. The ten depth layers are channels. The input is the field divided by its root-mean-square (rms) amplitude s, a channel holding $\log_{10} s$ (so the network knows how far the field is from saturation), a 16-dimensional embedding of the material numbers (a bias-free two-layer network) and two coordinate channels. A pointwise layer lifts these to 32 channels, two down-sampling blocks reduce the grid from 64^2 to 32^2 and 16^2 (64 and 128 channels), a middle block works at 16^2, and two up-sampling blocks return to 64^2 with skip connections from the matching resolution. A final block and a two-layer pointwise head give ten output channels, which are multiplied back by s. The U-shape lets the correction depend on both the local structure of the walls and the larger domain pattern. The model has 9.5M parameters and keeps only the lowest Fourier modes (up to 16 per direction), which costs nothing during incubation, because the Fourier SSM in Equation (6) acts on every mode.

Next-step training with a k-step rollout loss. Duhamel-NO is a next-step model: it maps frame P^n to frame P^{n+1} and is applied to its own output at inference. Single-step training would let errors compound at inference. We therefore train on windows of $k+1 = 5$ consecutive frames. Starting from the first frame, the model is rolled out for $k = 4$ steps, and the relative error of every step is summed:

$$\mathcal{L} = \frac{1}{k}\sum_{q=1}^{k} \frac{\|\tilde{P}^q - P^q\|^2}{\|P^q\|^2}, \qquad \tilde{P}^0 = P^0, \quad \tilde{P}^q = \Phi_\theta\big(\tilde{P}^{q-1}\big), \tag{9}$$

where Φ_θ is one step of Equation (6). Normalizing by $\|P^q\|$ matters here, because the early fields are about 10^3 times smaller than the saturated ones, and a plain squared error would ignore them. Unlike the pushforward trick [Brandstetter et al., 2022], which stops gradients through the model's

Table 1: **Accuracy on unseen materials and unseen noise** (mean over training seeds; **best**, second). D1: 256 held-out materials. D2: 300 independent materials, partly outside the training range. Correlation and relative error at frame 400, and relative error at frame 1. Poseidon-T is the best of the generative and pretrained models.

model	D1 corr.	D1 error	D2 error	frame-1 error
Duhamel-NO (ours)	**0.982**	**0.129**	**0.145**	**0.082**
F-FNO	0.974	0.167	0.195	0.215
U-Net	0.966	0.207	0.242	0.313
UNO	0.962	0.219	0.238	0.315
CNextU-Net	0.959	0.229	0.261	0.347
CNO	0.951	0.256	0.292	0.451
TFNO	0.949	0.272	0.265	0.564
FNO	0.947	0.270	0.268	0.568
Poseidon-T (pretrained)	0.553	2.917	2.570	4.342

own inputs, we backpropagate through all k steps. Because the pattern is decided early, 60% of the windows start in the first 120 frames and the rest start anywhere. We also apply random exact symmetries (90° rotations, reflections and $P_z \to -P_z$) to every window. Every baseline in Section 5 is trained with this same setup; optimizer settings are in Section D. Algorithm 1 summarizes the rollout from the initial noise (FFT, IFFT: fast Fourier transform and inverse).

5 Results

Baselines. We compare against seven neural operators and U-Nets: the Fourier neural operator (FNO) [Li et al., 2021] and its tensorized (TFNO) [Kossaifi et al., 2024] and factorized (F-FNO) [Tran et al., 2023] variants, UNO, U-Net, ConvNeXt U-Net (CNextU-Net) [Gupta and Brandstetter, 2023, Ohana et al., 2024], and the convolutional neural operator (CNO) [Raonić et al., 2023]. To make the comparison fair, every baseline is trained with the same setup as Duhamel-NO (material units, amplitude-aware input, the loss of Equation (9), augmentation), and its hyperparameters are tuned on the validation materials. This setup matters for the baselines: with standard training instead, no backbone exceeds correlation 0.84, at a relative error of 1.0–2.9, which is no better than predicting zero. Generative models (PDE-Refiner, diffusion, flow matching) and a fine-tuned foundation model, Poseidon-T [Herde et al., 2024], did not learn the task (Table 1).

5.1 Zero-shot generalization across materials and noise

Table 1 gives the results. On the 256 held-out D1 materials, Duhamel-NO reaches pattern correlation 0.982 and relative error 0.129 at frame 400 (all metrics with standard deviations in Section E). Its error is 23% lower than that of the strongest baseline, F-FNO. On D2, which it never saw and which is partly outside the training range, its error is 0.145, 26% lower than F-FNO. Three training seeds agree to within ± 0.001 in correlation.

The model does not lose accuracy when it extrapolates. Its error is 0.120 on held-out materials inside the sampled box and 0.133 on those at its edges. Across the five D2 regions, its correlation stays between 0.974 and 0.985: it is 0.985 on materials whose P_{S} lies up to $2.2\times$ above the training range, and 0.974 on first-order wells with r beyond the training range, which have a different free-energy landscape. Figure 2 shows a typical held-out material and a first-order well.

The comparison with UNO isolates the effect of the architecture. UNO uses the same network as Duhamel-NO's correction, with the same training, but without the Duhamel structure. Its error is 0.219 at frame 400 and 0.315 at frame 1, against 0.129 and 0.082 for Duhamel-NO. Building the Duhamel step into the operator, rather than asking the network to learn the whole update, accounts for this difference.

Why the error falls during incubation. Figure 3a,b shows the error at every frame. For every model it falls over the first ten frames and only then rises, although errors in an autoregressive rollout usually compound and no corrector is used. During incubation the dynamics act as the corrector: on the decaying Fourier modes, which are most of them, they shrink an error at every step, and on

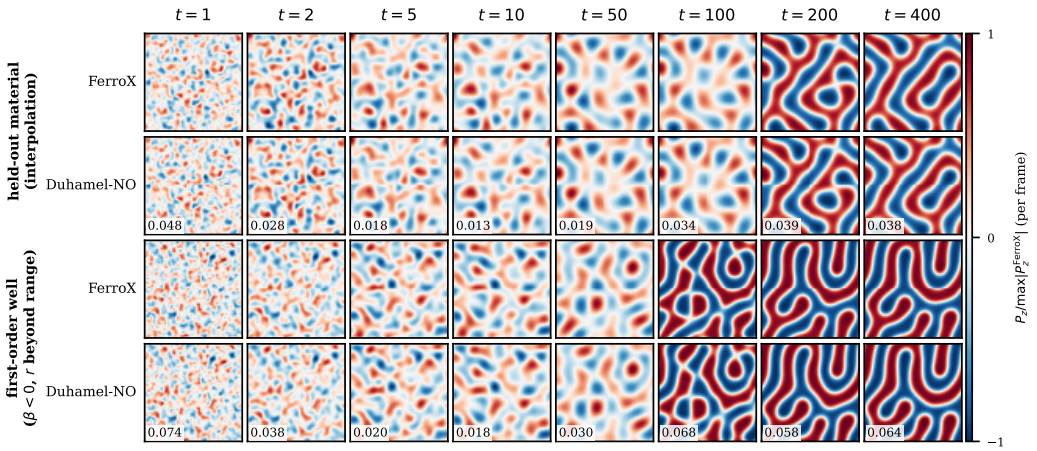


Figure 2: **Duhamel-NO rolled out from unseen noise on two test materials**, against FERROX (top layer; each frame scaled by the maximum of the true field; numbers: relative error). Each material is the median-error trajectory of its group (D1 interior; D2 first-order wells); mean errors are higher because of a tail of harder materials (Table 1).

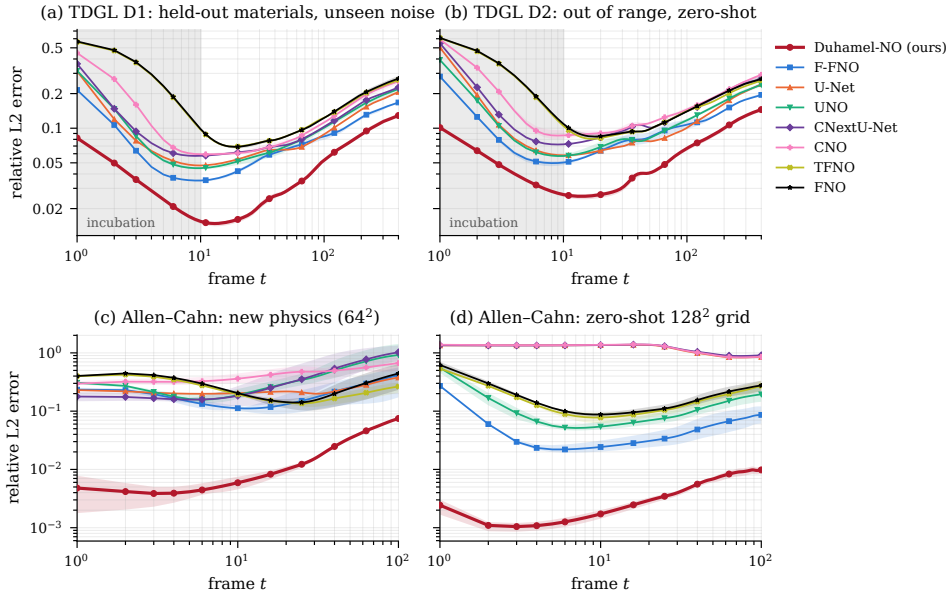


Figure 3: **Relative error at every frame**, log scales. (a,b) TDGL, D1 held-out materials and D2. (c,d) Allen–Cahn, new physics and zero-shot 128^2 grid (trained on 64^2). Mean over trajectories; bands: range over seeds.

the few growing modes an error grows only as fast as the field, so its relative size stays constant (Proposition 2). After nucleation this damping is lost, and errors slowly accumulate. Duhamel-NO propagates its errors with the exact factor $e^{\theta L}$, so it benefits most.

The gain of Duhamel-NO is largest at the start. At frame 1, its error is 0.082, against 0.215 for F-FNO and 0.568 for FNO, and the gap persists to frame 400. This is the behaviour Equation (6) predicts. An FNO keeps only a fixed set of low Fourier modes, and above that cut-off its linear response is a single constant. From the measured Fourier SSM, we compute that no model with the FNO's set of modes can exceed correlation 0.853 at frame 1 on these materials. The trained FNO reaches 0.826, just below this limit. Duhamel-NO applies the Fourier SSM to every mode and reaches 0.997.

5.2 Physical quantities

Table 2 lists the physical quantities of Section 3, recomputed from each model's prediction on the D1 held-out materials. In paired tests, Duhamel-NO is significantly better than the best baseline on 8 of 12 metrics and tied on the other 4, late-time morphology statistics (domain size, wall density, wall energy, P/P_{S} histogram). It predicts the nucleation frame exactly for 73% of trajectories (best baseline: 62%).

Table 2: **Errors in physical quantities**, D1 held-out materials (mean over trajectories and seeds; one seed for U-Net, CNextU-Net, TFNO; **best**, second). t_{nuc} in frames; $F(t)$ relative to the total energy drop; others: relative error at frame 400.

model	t_{nuc}	P_{S}	domain size	wall density	P/P_{S} hist.	φ	E_z	$F(t)$
Duhamel-NO (ours)	**0.28**	**0.003**	**0.005**	**0.007**	**0.010**	**0.139**	**0.135**	**0.004**
F-FNO	0.70	0.006	**0.005**	0.008	**0.010**	0.165	0.164	0.007
U-Net	0.61	0.008	**0.005**	0.008	0.011	0.201	0.200	0.006
UNO	0.54	0.024	0.006	0.009	0.020	0.220	0.218	0.012
CNextU-Net	0.55	0.005	0.006	0.008	**0.010**	0.224	0.222	0.007
CNO	0.46	0.006	0.008	0.010	0.011	0.256	0.253	0.008
TFNO	0.69	0.011	0.008	0.011	0.013	0.263	0.260	0.008
FNO	0.68	0.012	0.008	0.011	0.014	0.270	0.267	0.009

5.3 Speed

Timing one 400-frame trajectory on one dedicated A100 GPU, FERROX needs a median of 196 s per D1 trajectory (range 147–318 s). Duhamel-NO needs 1.66 s alone and 0.11 s per trajectory in batches of 16, about $118\times$ and $1814\times$ faster, at about the speed of the neural baselines: the Fourier SSM adds almost nothing to the cost of the network. Training (682 runs, about 37 GPU-hours) pays for itself after about 700 trajectories.

5.4 Allen–Cahn: new physics and new grids

No measured Fourier SSM exists here, so Duhamel-NO learns its linear symbol as a polynomial in the dimensionless wavenumber $k\mu$, which does not depend on the grid. In range, its final error is 0.012 (best baseline 0.043, F-FNO). On new physics (Figure 3c) it reaches 0.075, $3.5\times$ lower than the best baseline (0.265, TFNO). Trained on 64^2, it runs unchanged on 128^2 (Figure 3d) with error 0.010, while F-FNO and FNO degrade to 0.086 and 0.276. This is resolution transfer: the linear step is a function of the physical wavenumber $k\mu$, not of the grid index, so on the finer grid the same trained model advances every mode, including modes that the training grid cannot represent, by its exact linear factor.

6 Conclusion

Duhamel-NO is built from the TDGL equation through Duhamel's principle: the Fourier SSM, which decides the domain pattern, is computed in closed form, and only the nonlinear correction that saturates the growth is learned. Because β and γ do not enter the Fourier SSM, it is set by α alone, and the network conditions on the material through a few closed-form numbers. It generalizes zero-shot to unseen noise, held-out and out-of-range materials and first-order wells, with lower error and better physical quantities than seven tuned baselines, about $1814\times$ faster than the solver.

Limitations. Duhamel-NO is built for one class of problem, which we call *noise-seeded dynamics*: a small random initial field (here about 0.5% of the final amplitude) is amplified by an unstable linear stage, and the pattern that emerges is decided there. It needs an accurate linear operator, which we measure from the first frames of the training simulations. With only 32 training materials, the interpolated Fourier SSM is too coarse in α, and F-FNO is more accurate (0.186 vs. 0.239). Finally, all ferroelectric data come from one simulator, a uniaxial model without elastic coupling, with one electrode and dielectric configuration, and are compared with simulation rather than experiment.

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A Derivations and proofs

Throughout, $\hat{f}(k)$ is the in-plane Fourier transform of a field f, $k = (k_x, k_y)$, and a field on the $64 \times 64 \times 10$ grid is a vector of ten depth layers at every in-plane point. For a field u we write $\|u\|$ for the L_2 norm over all voxels.

A.1 The split of the TDGL equation

The free energy of Equation (1) is $F = F_2 + F_{46}$, with

$$F_2[P_z] = \int \Big[\tfrac{\alpha}{2} P_z^2 + \tfrac{G_{11}}{2}\|\nabla P_z\|^2\Big] d\mathbf{r} + U_{\text{elec}}[P_z], \qquad F_{46}[P_z] = \int \Big[\tfrac{\beta}{4} P_z^4 + \tfrac{\gamma}{6} P_z^6\Big] d\mathbf{r}.$$

The electrostatic energy is $U_{\text{elec}} = -\frac{1}{2}\int E_z[P_z]\, P_z \, d\mathbf{r}$, where $E_z[P_z]$ is the z-component of the field that solves the Poisson problem of the stack with P_z as its source. The Poisson problem is linear with fixed (grounded) boundary values, so $E_z[P_z]$ is linear in P_z and the map $P_z \mapsto -E_z[P_z]$ is symmetric. The variational derivatives are therefore

$$\frac{\delta F_2}{\delta P_z} = \alpha P_z - G_{11}\nabla^2 P_z - E_z[P_z], \qquad \frac{\delta F_{46}}{\delta P_z} = \beta P_z^3 + \gamma P_z^5,$$

and $\partial_t P_z = -\Gamma\, \delta F/\delta P_z$ is exactly $\partial_t P_z = \mathcal{L} P_z + \mathcal{N}(P_z)$ with $\mathcal{L}$ and $\mathcal{N}$ as in Equation (2). $\mathcal{L}$ is linear and self-adjoint (up to the factor $-\Gamma$), and it contains every coefficient that survives as $P_z \to 0$: α, G_{11} and the geometry of the stack. $\mathcal{N}$ contains only β and γ. This is why, in Duhamel-NO, the Fourier SSM depends on α alone, while β and γ enter only the correction (whose network also receives α through the material numbers).

A.2 The effective force, the exponential-Euler step and the filter φ_1

Section 4.2 shows that $\tau\varphi_1(\tau\mathcal{L})$ is symmetric positive definite, so the effective force of Equation (5) is well defined, and that it equals $\int_0^\tau e^{(\tau-s)\mathcal{L}} ds$. Below we also use that $\varphi_1(z) = \int_0^1 e^{zr} dr$ is increasing on the real line, and that, by Equation (4), $\tau\varphi_1(\tau\mathcal{L}) N_{\text{eff}}(P_z(t)) = \int_0^\tau e^{(\tau-s)\mathcal{L}} \mathcal{N}(P_z(t+s))\, ds$.

Distance from the frozen force. Freezing the integrand of Equation (4) at $s = 0$ gives

$$P_z(t+\tau) \approx e^{\tau\mathcal{L}} P_z(t) + \tau\, \varphi_1(\tau\mathcal{L})\, \mathcal{N}(P_z(t)),$$

the exponential-Euler scheme [Cox and Matthews, 2002, Hochbruck and Ostermann, 2010], so $N_{\text{eff}} = \mathcal{N}(P_z(t))$ if $\mathcal{N}$ is constant over the frame. In general,

$$\tau\varphi_1(\tau\mathcal{L})\big[N_{\text{eff}}(P_z(t)) - \mathcal{N}(P_z(t))\big] = \int_0^\tau e^{(\tau-s)\mathcal{L}} \big[\mathcal{N}(P_z(t+s)) - \mathcal{N}(P_z(t))\big] ds,$$

and the norm of the right-hand side is at most $\frac{\tau^2}{2} \sup_{s\le\tau}\|e^{(\tau-s)\mathcal{L}}\| \sup_{s\le\tau}\|\frac{d}{ds}\mathcal{N}(P_z(t+s))\|$. On a mode with $\tau\lambda \ge -1$, $\tau\varphi_1(\tau\lambda) \ge \tau\varphi_1(-1) = 0.63\,\tau$, so on such modes $N_{\text{eff}} - \mathcal{N}(P_z(t)) = O(\tau)$. It vanishes when $\mathcal{N}$ does not change within a frame, and in particular during incubation, where $\mathcal{N} = O(P_z^3)$. On stiff modes ($\tau\lambda \ll -1$), $\tau\varphi_1(\tau\lambda) \approx 1/|\lambda|$, so any component of N_{eff} on those modes enters the step with a small weight, and the network does not need to resolve it. Duhamel-NO replaces N_{eff} by a learned N_ψ, trained so that the whole step matches the data: it learns the effective force along the path, not the frozen value at $s = 0$.

A.3 Proof of Proposition 1

Proof of Proposition 1. Let $\Lambda(k)$ be diagonal and set $h^t(k) = V(k)^{-1}\hat{P_z}^t(k)$. Then $h^{t+1} = V^{-1}B\hat{P_z}^t = V^{-1}V\Lambda V^{-1}\hat{P_z}^t = \Lambda h^t$ and $\hat{P_z}^t = V h^t$. This is the state-space recurrence $h^{t+1} = Ah^t + Bx^t$, $y^t = Ch^t + Dx^t$ with $A = \Lambda$, $C = V$, $D = 0$ and an input x^t that is the initial noise at $t = 0$ and zero afterwards. Every mode k is an independent copy of this system, so the full linear dynamics are a bank of diagonal, linear, time-invariant SSMs, one per in-plane mode. □

Remark. Diagonalizability holds generically and is verified numerically: on the reference material the relative reconstruction error $\|V\Lambda V^{-1} - B\|_\infty / \|B\|_\infty$ is 2.9×10^{-15}, with condition number $\kappa(V) \le 6.39$. In the continuum the generator is self-adjoint up to the mobility (Section A.1), so its eigenvalues are real; in the fitted discrete map all 128 unstable eigenvalues of the reference material are real to 3.9×10^{-8}, and complex pairs occur only among modes that decay within two frames.

A.4 Material units and the shared nonlinearity

Write $P_z = P_S v$ and measure time in units of λ. Divide $\mathcal{N}$ by $\Gamma|a_{\text{sat}}|P_S$, where $|a_{\text{sat}}| = \beta P_S^2 + \gamma P_S^4$ defines P_S (Section C.3):

$$\frac{\mathcal{N}(P_S v)}{\Gamma|a_{\text{sat}}|P_S} = -\frac{\beta P_S^2}{|a_{\text{sat}}|}v^3 - \frac{\gamma P_S^4}{|a_{\text{sat}}|}v^5 = -(1-r)\,v^3 - r\,v^5, \qquad r = \frac{\gamma P_S^4}{|a_{\text{sat}}|}.$$

The second equality uses $\beta P_S^2/|a_{\text{sat}}| = 1 - r$, which follows from the definition of P_S. So in material units every material has the same nonlinearity up to the single number r, and the corresponding reduced potential $\phi_r(v) = -v^2/2 + (1-r)v^4/4 + rv^6/6$ has its minima at $v = \pm 1$ for every r (Figure 6). $r \in [0, 1)$ is a second-order well, $r = 1$ means $\beta = 0$, and $r > 1$ is a first-order well ($\beta < 0$), held only by the sextic term.

A.5 Why the error falls during incubation

The following statement makes the argument of Section 5.1 precise for the linear stage. Let the true field obey $P_{n+1} = BP_n$ with B diagonalizable, $B = V\Lambda V^{-1}$, and split the eigen-coordinates into an unstable set $\mathcal{U}$ ($|\lambda_i| > 1$) and a stable set $\mathcal{S}$ ($|\lambda_i| \le \rho_s < 1$). Let a surrogate produce $\tilde{P}_{n+1} = B\tilde{P}_n + \delta_n$, where δ_n is the error it adds at step n.

Proposition 2 (error propagation in the linear stage)**.** The error $e_n = \tilde{P}_n - P_n$ satisfies $e_n = B^n e_0 + \sum_{j<n} B^{n-1-j}\delta_j$. In eigen-coordinates $c = V^{-1}e$, the stable components of the carried-over error decay geometrically, $|c_i(B^n e_0)| \le \rho_s^n |c_i(e_0)|$ for $i \in \mathcal{S}$, while the unstable components grow at exactly the rate of the corresponding components of the true field. Consequently, if the injected errors are small after the first step ($\|\delta_j\| \ll \|e_0\|$ for $j \ge 1$), the relative error $\|e_n\|/\|P_n\|$ decreases with n as long as the stable part of e_0 dominates the error and the field is dominated by its unstable part.

Proof. The recursion follows by induction. In eigen-coordinates B^n acts as Λ^n, so component i of $B^n e_0$ is $\lambda_i^n c_i(e_0)$: it shrinks as ρ_s^n on $\mathcal{S}$ and grows as $|\lambda_i|^n$ on $\mathcal{U}$, the same factor that multiplies component i of $P_n = B^n P_0$. The ratio of unstable error to unstable field is therefore constant, while the stable error decays and the stable part of the field decays too, but the field norm is increasingly carried by $\mathcal{U}$. □

For Duhamel-NO in the linear stage, the hypotheses of Proposition 2 hold with the true rates: the Fourier SSM of Equation (6) is the measured B, and the correction is $O(\|v\|)$ times a small learned term.

A.6 The linear stage: an exact factorization and the truncation cap

These statements explain why truncated neural operators are limited in the first frames (Section 5.1). Write the one-frame map as $S(u) = Au + R(u)$ with A a Fourier multiplier $g(k)$ (a $Z \times Z$ matrix per wavenumber) and $\|R(u)\| \le C\|u\|^p$, $p \ge 2$. A field F with its spatial mean removed is written $\tilde{F}$, and $\rho(F,T) = \mathbb{E}\langle \tilde{F}, \tilde{T}\rangle / \sqrt{\mathbb{E}\|\tilde{F}\|^2\,\mathbb{E}\|\tilde{T}\|^2}$ is the population pattern correlation. Let $J_1 F$ be the orthogonal projection of the random field F onto the linear span of the starting-noise components, i.e. the part of F that is linear in the noise (for Gaussian noise, the first Wiener chaos [Nualart, 2006]; Gaussianity is not needed). Let $\mathcal{H}(F) = 1 - \mathbb{E}\|J_1\tilde{F}\|^2/\mathbb{E}\|\tilde{F}\|^2$ be the share of the output that is not linear in the noise, and $\rho_{\text{lin}}(F,T) = \rho(J_1 F, T)$.

Proposition 3 (exact factorization)**.** Let $T = A\epsilon\xi$ be the true first frame from noise $\epsilon\xi$. For every surrogate output $G = G(\epsilon\xi)$ with finite second moment, $\rho(G,T) = \rho_{\text{lin}}(G,T)\sqrt{1-\mathcal{H}(G)} \le \sqrt{1-\mathcal{H}(G)}$.

Proof. $\tilde{T}$ lies in the linear span of the noise components and $\tilde{G} - J_1\tilde{G}$ is orthogonal to that span, so $\mathbb{E}\langle \tilde{G}, \tilde{T}\rangle = \mathbb{E}\langle J_1\tilde{G}, \tilde{T}\rangle$ and $\mathbb{E}\|\tilde{G}\|^2 = \mathbb{E}\|J_1\tilde{G}\|^2 + \mathbb{E}\|\tilde{G} - J_1\tilde{G}\|^2$. Hence $\rho = \rho_{\text{lin}}\sqrt{\mathbb{E}\|J_1\tilde{G}\|^2/\mathbb{E}\|\tilde{G}\|^2} = \rho_{\text{lin}}\sqrt{1-\mathcal{H}}$, and $|\rho_{\text{lin}}| \le 1$ by Cauchy–Schwarz. □

Proposition 4 (three behaviours as the noise shrinks)**.** (a) If G is Lipschitz with a non-zero, spatially varying output at zero input (biases with positional channels), then $\mathcal{H}(G(\epsilon\xi)) \to 1$ and $\rho \to 0$ as

$\epsilon \to 0$. (b) If $G(u) = r(u)\Phi(u/r(u), \log r(u))$ with $r = \mathrm{rms}(u)$, then $\mathcal{H}$ depends on ϵ only through the explicit $\log r$ input. (c) If $G(u) = \hat{A}u + N(u)$ with $\|N(u)\| \le K\|u\|^2$, then $\mathcal{H} = O(\epsilon^2)$ and $\rho \to \rho_{\mathrm{lin}}(\hat{A}, A)$.

Proof. (a) Let L be the Lipschitz constant, $b = \tilde{G}(0) \neq 0$ and $c = L\epsilon(\mathbb{E}\|\xi\|^2)^{1/2}$. Then $\mathbb{E}\|\tilde{G}(\epsilon\xi) - b\|^2 \le c^2$, so the mean part satisfies $\|J_0\tilde{G}\| \ge \|b\| - c$ and $\mathbb{E}\|\tilde{G}\|^2 \le (\|b\| + c)^2$, giving $\mathcal{H} \ge (\|b\| - c)^2/(\|b\| + c)^2 \to 1$; and $\mathbb{E}\|J_1\tilde{G}\|^2 \le c^2$ bounds $\rho \le c/(\|b\| - c) \to 0$. (b) $\mathcal{H}$ is invariant under deterministic rescaling, so $\mathcal{H}(G(\epsilon\xi)) = \mathcal{H}(r(\xi)\Phi(\xi/r(\xi), \log\epsilon + \log r(\xi)))$. (c) $\hat{A}\epsilon\xi$ is linear in the noise, so the non-linear share comes only from N, with $\mathbb{E}\|N\|^2 \le K^2\epsilon^4\mathbb{E}\|\xi\|^4$, while $\mathbb{E}\|\tilde{G}\|^2 \ge (\epsilon\|\hat{A}\tilde{\xi}\| - K\epsilon^2(\mathbb{E}\|\xi\|^4)^{1/2})^2$. Hence $\mathcal{H} = O(\epsilon^2)$, and the linear part of N is $O(\epsilon^2)$, so $\rho_{\mathrm{lin}}(G) \to \rho_{\mathrm{lin}}(\hat{A})$. □

Proposition 5 (FNO linearization)**.** For an FNO without positional channels, the Jacobian at any spatially homogeneous state is a Fourier multiplier $M(k)$ that equals one fixed matrix M_∞ for every k outside the retained mode set $\mathcal{K}$.

Proof. At a homogeneous state every pre-activation is spatially constant, so layer l has Jacobian $D_l(W_l + R_l(k)\mathbb{1}[k \in \mathcal{K}])$ with $D_l = \mathrm{diag}(\phi'(\cdot))$. Products of multipliers are multipliers, and outside $\mathcal{K}$ only the k-independent W_l terms remain. With coordinate channels the action above the cut-off is multiplication by a fixed smooth field, so the statement holds up to the low-wavenumber content of that field. □

Proposition 6 (truncation cap)**.** If the linear response of G is constant outside $\mathcal{K}$, then for white noise $\rho_{\mathrm{lin}}(G, A^n) \le \rho_{\mathcal{K}}(n) = \left(1 - |\mathcal{K}^c|\,\mathrm{Var}_{k\in\mathcal{K}^c}[g_n(k)]/\sum_{k\neq 0}|g_n(k)|^2\right)^{1/2}$, and $\rho(G, T_n) \le \rho_{\mathcal{K}}(n)\sqrt{1 - \mathcal{H}(G)}$.

Proof. For translation-equivariant G and white noise, $J_1G = \mathcal{F}^{-1}[m(k)\mathcal{F}\xi]$ (linear regression on white noise is a convolution), so ρ_{lin} is the spectral cosine between m and g_n. Maximizing it over m with $m \equiv c$ on $\mathcal{K}^c$ gives $m = g_n$ on $\mathcal{K}$ and c equal to the mean of g_n on $\mathcal{K}^c$; substituting gives $\rho_{\mathcal{K}}(n)$. The second bound follows from Proposition 3. □

By Proposition 5, an FNO's linear response is constant outside its mode set, so Proposition 6 applies to it. Proposition 6 is what the main text evaluates. Computed from the measured Fourier SSM of the D1 held-out materials, the cap for the FNO used here (modes up to 8 in each in-plane direction) is 0.853, 0.882 and 0.926 at frames 1–3 and 0.996 at frame 10. The trained FNO reaches 0.826, 0.877 and 0.923 (Figure 4). For rms-normalized networks (class (b)) the cap is an empirical statement, not a proven bound; it does not apply to architectures with a different mode set (F-FNO) or without spectral truncation (U-Nets, CNO). Duhamel-NO is of class (c) of Proposition 4 in its linear part: its Fourier SSM acts on every mode, so no cap applies, and it reaches 0.997 at frame 1.

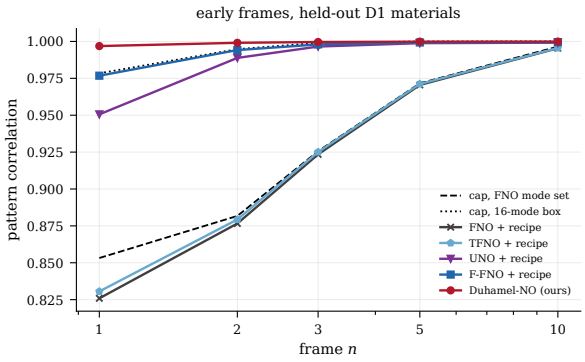


Figure 4: **Truncation cap against trained models** (D1 held-out materials, frames 1–10). Dashed and dotted: the best correlation any model with the FNO's mode set, or with a 16-mode box, can reach (Proposition 6), computed from the measured Fourier SSM. Markers: trained models.

B Algorithms

Algorithm 1 (Section 4) rolls out Duhamel-NO from the initial noise. The three procedures below measure the Fourier SSM (Algorithm 2), build it for a new material (Algorithm 3), and train the network (Algorithm 4).

Algorithm 2 Measuring the Fourier SSM of one training material

Require: frames $P^0, \dots, P^8$ of one training simulation (field still small); frame interval θ (in growth times)

$\hat{P}^t \leftarrow$ in-plane FFT of P^t for $t = 0, \dots, 8$ ▷ $\hat{P}^t(k) \in \mathbb{C}^{10}$, one vector per mode

for each radial bin b of $|k|$ (46 bins) **do**

  $X \leftarrow [\hat{P}^t(k)]_{k\in b,\ t<8}$, $Y \leftarrow [\hat{P}^{t+1}(k)]_{k\in b,\ t<8}$ ▷ snapshot pairs

  $B_b \leftarrow YX^{\mathsf{H}}(XX^{\mathsf{H}} + \epsilon I)^{-1}$ ▷ least squares (small ridge), 10×10

  $L_b \leftarrow \theta^{-1} \log B_b$ ▷ generator, symmetrized

bins whose modes decay below round-off within one frame get a learnable, strongly damped rate; the uniform mode is damped

return $\{L_b\}$ and $\log|\alpha|$ of this material

Algorithm 3 The Fourier SSM of a new material

Require: α of the new material; generators $\{L_b^{(i)}\}$ of all training materials i

$\mathcal{I} \leftarrow$ the eight training materials with $\log|\alpha_i|$ closest to $\log|\alpha|$

$L_b \leftarrow \frac{1}{8}\sum_{i\in\mathcal{I}} L_b^{(i)}$ for every bin b ▷ β, γ not needed: they do not enter $\mathcal{L}$

return $L(k) = L_{b(|k|)}$, and $e^{\theta L(k)}$, $\theta\varphi_1(\theta L(k))$ by `matrix_exp`

Algorithm 4 Training Duhamel-NO

Require: training trajectories (682 materials, one trajectory each); Fourier SSMs from Algorithm 2

initialize N_ψ with a zero read-out ▷ the untrained model is the exact linear solution

for 20 000 steps **do**

  sample 16 windows of $k{+}1 = 5$ frames; with probability 0.6 the window starts in frames 0–120 (growth phase)

  apply a random exact symmetry: 90° rotation, reflection, $P_z \to -P_z$

  roll out $k = 4$ steps with Algorithm 1 from the first frame of the window (gradients flow through all steps)

  $\mathcal{L} \leftarrow \frac{1}{k}\sum_{q=1}^{k} \|\tilde{P}^q - P^q\|^2 / \|P^q\|^2$ ▷ relative L_2

  AdamW (Adam with decoupled weight decay) step (one-cycle schedule, weight decay 10^{-4}, gradient clipping at 1)

C Datasets and material units

C.1 Simulations

All data are FERROX [Kumar et al., 2023] simulations of the stack of Section 3: explicit-Euler TDGL steps with a Poisson solve for the whole stack at every step. Each run starts from a field drawn uniformly from $\pm 2 \times 10^{-3}$ C/m^2 in every cell, set by an integer seed: D1 material i uses seed $2000 + i$, and D2 uses $6000 + i$. A trajectory is rejected if it has non-finite values or $|P_z| > 1.35 P_S$; none was rejected.

Table 3: **Datasets.** Every trajectory has its own noise seed; no seed is shared between training and test data. Hysteresis is described in Section F.

dataset	grid	frames	varied parameters	training	test
D1	$64^2 \times 10$	401	α, β, γ (Latin hypercube in $\log|\alpha|, P_S, r$)	682 (+63 validation)	256 held-out materials (82 interior, 174 edge)
D2	$64^2 \times 10$	401	α, β, γ drawn independently, five regions	–	300 (5×60), no retraining
Hysteresis	$64^2 \times 10$	440–2765	α, β, γ, peak voltage	90 loops (+10 validation)	30 held-out material, 34 new noise, 10 closed set

Table 3 summarizes the datasets, and Figure 5 shows where the D1 and D2 materials lie in $(|\alpha|, P_S, r)$.

D1. The coefficients are $\gamma = r|a_{sat}|/P_S^4$ and $\beta = (1-r)|a_{sat}|/P_S^2$, so every sample is a valid double well. Sampling the raw coefficients instead would reject more than 95% of draws, because the admissible range of β moves by two orders of magnitude across the range of α. The validation materials are every twelfth training material.

D2. 300 materials whose α, β, γ are drawn independently (log-uniform) and assigned to five regions of 60: **A** small P_S; $\mathbf{B}_{\text{units}}$ large P_S, up to 2.2× the training maximum (a pure change of units: the dimensionless numbers stay in range); $\mathbf{B}_\alpha$ large $|\alpha|$ with small P_S; **C** first-order wells with $r = 1.15$–2 (a change of a dimensionless number, i.e. new physics); **D** all of these together.

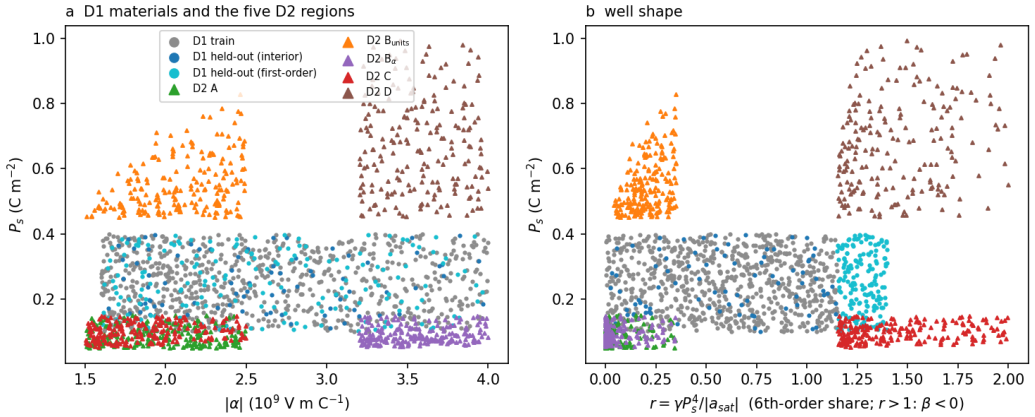


Figure 5: **Where the materials lie.** D1 materials (circles; grey: training, blue: held out) and D2 materials (triangles, coloured by region) in well depth $|\alpha|$, saturation polarization P_S and well shape r. D2 leaves the training range in P_S and in r.

C.2 Free-energy landscapes

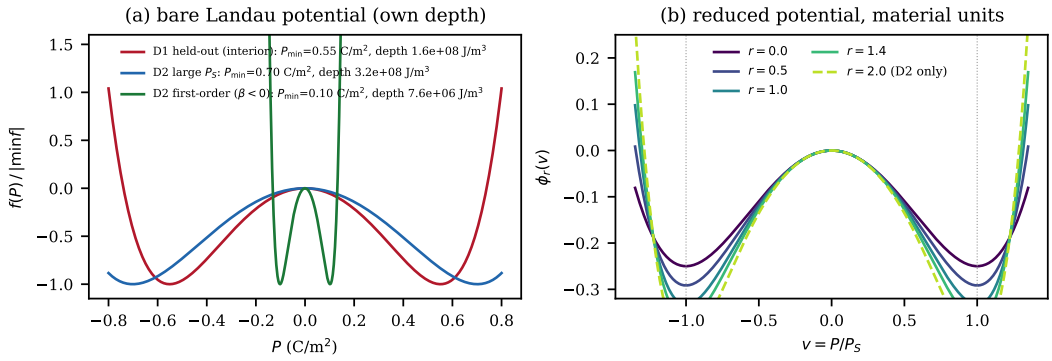


Figure 6: **Free-energy landscapes of the test materials.** (a) Bare Landau potential $f(P) = \alpha P^2/2 + \beta P^4/4 + \gamma P^6/6$ of the three materials of Figure 2 and the D2 large-P_S material, each divided by its own well depth: the wells differ in position by a factor of seven and in depth by two orders of magnitude, and the first-order material ($\beta < 0$) has a narrow, steep well. (b) In material units every material has the reduced potential $\phi_r(v)$ of Section A.4, with minima at $v = \pm 1$; only its shape, set by r, differs. D2 reaches $r = 2$, beyond the training range (dashed).

C.3 Material units

The depolarization field of the dielectric renormalizes α, and it does so differently for the two things α controls: the growth of small fluctuations and the saturated amplitude. Fitting simulations gives

a growth coefficient $|a_{\mathrm{rate}}| \propto |\alpha| - 1.37 \times 10^9$, which sets the growth time λ, and a saturation coefficient $|a_{\mathrm{sat}}| = 1.389\,(|\alpha| - 1.31 \times 10^9\,(1 - 0.0775\,r))$ (units of α, V m/C). The saturated polarization P_{S} solves $|a_{\mathrm{sat}}| = \beta P_{\mathrm{S}}^2 + \gamma P_{\mathrm{S}}^4$, and the shape number is $r = \gamma P_{\mathrm{S}}^4 / |a_{\mathrm{sat}}|$. This law predicts P_{S} of the D1 materials to 1.03% on average (3.41% at worst) and of the D2 materials, whose P_{S} lies up to $2.2\times$ above the fit range, to 1.9% (Figure 7). The value 1.37×10^9 is a critical point: for $|\alpha|$ below it the depolarization field outweighs the Landau driving force and the film never nucleates; all D1 and D2 materials lie above it. Over 120 D1 materials, the change of units also reduces the coefficient of variation of the settled amplitude from 0.38 to 0.083.

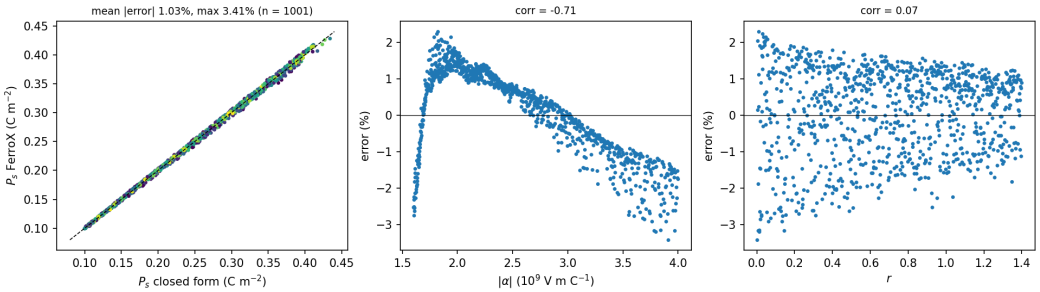


Figure 7: **Closed-form saturation polarization** against the value measured in FERROX (99.5% quantile of $|P|$ over the settled frames) for all D1 materials, coloured by r, with residuals against $|\alpha|$ and r.

D Training and evaluation details

Training. Every model: 20 000 optimizer steps, batch 16, $k = 4$ unrolled steps per loss evaluation, AdamW (weight decay 10^{-4}) with a one-cycle schedule (5% warm-up), gradient clipping at 1, 32-bit floating point (fp32), and exact symmetry augmentation. Generative models are trained on their own objectives (one-step denoising, or flow matching on the residual) and sampled with 20 denoising diffusion implicit model (DDIM) steps (autoregressive conditional diffusion model, ACDM), 8 Euler steps (flow matching) or 3 refinement steps (PDE-Refiner). Poseidon-T is fine-tuned from its public pretrained weights for 20 000 steps.

Without our training setup. Every baseline is trained with the setup of Section 4.4. Without it, a baseline receives the raw material parameters (standardized) as constant input channels and periodic coordinate channels. Table 4 shows that every backbone needs the setup.

Table 4: **Standard training against our training setup**, same backbone, same (4000-step) tuning budget, FerroX validation split (new materials, new noise). The best standard configuration is a relative-L_2 loss with per-sample normalization for every backbone except Transolver* (relative-L_2 loss alone).

	best standard configuration		with our training setup	
backbone	corr@400	rel. L2@400	corr@400	rel. L2@400
F-FNO	0.837	2.92	0.895	0.41
U-Net	0.841	2.04	0.909	0.37
UNO	0.842	1.87	0.863	0.47
CNextU-Net	0.742	1.52	0.853	0.49
CNO	0.622	1.00	0.907	0.38
TFNO	0.678	2.18	0.863	0.50
FNO	0.819	1.79	0.849	0.52
Transolver*	0.044	1.76	0.066	1.30

Selection. Learning rate and configuration are chosen per model on the validation split (Table 5 lists the result for every model and task), by correlation minus 0.1 times the relative error (for hysteresis, the loop error). Variants of U-Net, CNextU-Net and CNO without coordinate channels (and CNO

with batch normalization, BatchNorm) are admitted as candidates. The chosen configuration is retrained with three seeds (one for the slowest models).

Evaluation choices. Predictions are clamped to three times the largest training value of the field, scaled per sample by its amplitude unit, so that out-of-range materials are not cut off. The neuralop FNO with `n_modes=(16,16)` keeps modes up to 8 in each direction; the truncation cap (Proposition 6) uses this true mode set. Timing uses one A100 with nothing else running.

Reproducibility. Every number in the paper and in this appendix is written as a LaTeX macro or a generated table from the benchmark result files (`make_duhamel_macros.py`, `make_physics_table.py`, `appendix/copy_assets.py`); none is typed by hand. The rollout figures are produced from the evaluated checkpoints by `make_rollout_fig.py` and `appendix/gallery_plot.py`.

Table 5: **Validation-selected configuration, size, number of seeds and training time** (one A100) of every model on every task; MSE: mean squared error loss.

task	model	params (M)	seeds	train (min)	lr	configuration
FerroX TDGL	Duhamel-NO (ours)	9.49	3	39	0.0003	our setup
	F-FNO	1.19	3	42	0.001	our setup
	U-Net	10.95	1	76	0.0003	our setup
	UNO	9.49	3	27	0.001	our setup
	CNextU-Net	9.93	3	59	0.0003	our setup
	CNO	5.50	3	93	0.001	our setup
	TFNO	1.03	3	27	0.001	our setup
	FNO	2.42	3	27	0.001	our setup
	Transolver	2.19	1	161	0.001	our setup
	Poseidon-T	20.79	1	129	0.0001	rel. L2
	U-Net + ACDM diffusion	10.94	1	16	0.001	global noise scale
	U-Net + flow matching	10.94	1	23	0.001	global noise scale
	U-Net + PDE-Refiner	10.94	1	25	0.001	global noise scale
hysteresis	Duhamel-NO (ours)	9.49	3	28	0.001	our setup
	F-FNO	1.19	1	32	0.001	our setup
	U-Net	10.94	1	79	0.0003	MSE
	UNO	9.49	3	19	0.001	our setup
	CNextU-Net	9.93	1	78	0.0003	our setup
	CNO	5.50	3	106	0.0003	our setup
	TFNO	1.03	1	20	0.001	MSE
	FNO	2.42	3	21	0.001	our setup
	Transolver	2.18	3	191	0.0003	MSE
	Poseidon-T	20.79	1	115	0.0001	rel. L2
	U-Net + flow matching	10.95	1	21	0.0003	per-sample noise scale
	U-Net + PDE-Refiner	10.95	1	22	0.0003	global noise scale

E Additional results on ferroelectric domain formation

Table 6 gives every model with standard deviations over seeds, and Table 7 and Figure 8 break the results down by test subset: Duhamel-NO is the most accurate model on every subset, inside and outside the training range.

Table 6: **All models** (mean ± std over seeds; **best**). Pattern correlation and relative L_2 error at frame 400 on D1 (held-out materials) and D2 (out of range), and relative error at frame 1 (D1).

			D1 held-out (256)		D2 out-of-range (300)		frame 1
model	family	seeds	corr@400	rel. L2@400	corr@400	rel. L2@400	rel. L2
Duhamel-NO (ours)	ours	3	**0.982**±**0.001**	**0.129**±**0.003**	**0.978**±**0.001**	**0.145**±**0.004**	**0.082**±**0.000**
F-FNO	spectral NO	3	0.974±0.001	0.167±0.003	0.968±0.000	0.195±0.004	0.215±0.001
U-Net	U-Net	1	0.966	0.207	0.954	0.242	0.313
UNO	spectral NO	3	0.962±0.003	0.219±0.012	0.959±0.002	0.238±0.008	0.315±0.005
CNextU-Net	U-Net	3	0.959±0.000	0.229±0.002	0.949±0.002	0.261±0.010	0.347±0.018
CNO	conv. NO	3	0.951±0.004	0.256±0.011	0.937±0.004	0.292±0.005	0.451±0.003
TFNO	spectral NO	3	0.949±0.005	0.272±0.017	0.952±0.003	0.265±0.012	0.564±0.004
FNO	spectral NO	3	0.947±0.004	0.270±0.015	0.947±0.003	0.268±0.012	0.568±0.003
Poseidon-T	foundation	1	0.553	2.917	0.414	2.570	4.342
U-Net + ACDM diffusion	diffusion	1	0.206	1.293	0.092	2.854	5.209
U-Net + flow matching	flow matching	1	0.169	1.330	0.094	1.895	5.210
U-Net + PDE-Refiner	diffusion	1	0.053	3.578	0.033	2.158	5.351
Transolver	transformer	1	0.003	1.329	0.008	1.313	0.757

Table 7: **Correlation at frame 400 per test subset** (mean ± std over seeds). D1: interpolation (inside the sampled box) and extrapolation (at its edges). D2 regions as in Section C.

model	D1 extrap.	D1 interp.	D2-A	D2-B_α	D2-B_{units}	D2-C	D2-D
Duhamel-NO (ours)	0.981±0.001	0.982±0.001	0.974±0.001	0.977±0.002	0.985±0.002	0.974±0.001	0.978±0.003
F-FNO	0.975±0.001	0.973±0.001	0.967±0.002	0.958±0.002	0.982±0.001	0.970±0.003	0.961±0.001
U-Net	0.963	0.972	0.967	0.970	0.979	0.941	0.915
UNO	0.959±0.004	0.970±0.002	0.958±0.010	0.949±0.003	0.976±0.004	0.951±0.008	0.960±0.003
CNextU-Net	0.958±0.000	0.962±0.001	0.951±0.008	0.957±0.004	0.969±0.002	0.941±0.003	0.926±0.016
CNO	0.949±0.004	0.955±0.007	0.917±0.008	0.952±0.011	0.959±0.004	0.923±0.018	0.934±0.009
TFNO	0.946±0.005	0.954±0.006	0.958±0.004	0.942±0.007	0.973±0.006	0.954±0.004	0.932±0.007
FNO	0.944±0.006	0.954±0.003	0.952±0.002	0.927±0.004	0.965±0.002	0.949±0.005	0.941±0.014

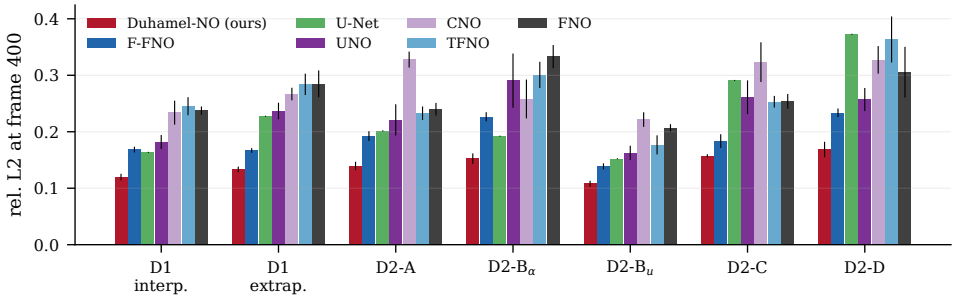


Figure 8: **Relative error at frame 400 per test subset** (bars: standard deviation over seeds). Duhamel-NO is the most accurate model on every subset, including those outside the training range.

E.1 Physical quantities

Figure 9 shows the physical metrics of Table 2 on D1 in more detail, and Figure 10 repeats them on D2, without retraining.

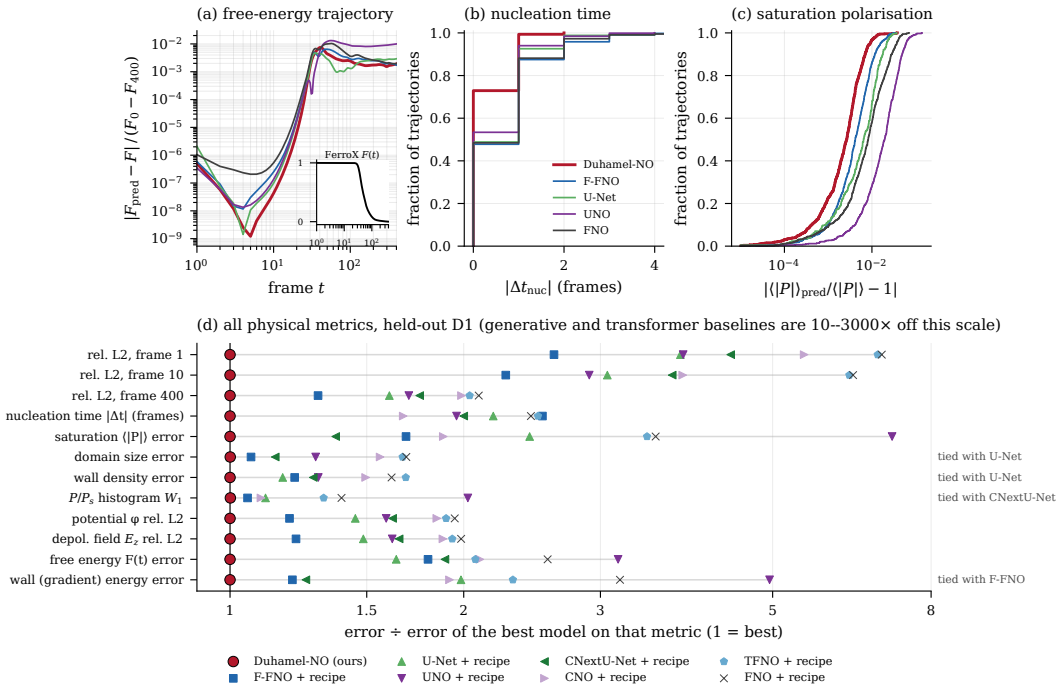


Figure 9: **Physical consistency on the D1 held-out materials.** (a) Error of the free-energy trajectory relative to the total energy drop (inset: the true $F(t)$). (b, c) Cumulative distributions of the nucleation-time error and of the saturation-polarization error (further left is better). (d) Every metric as the ratio of each model's error to that of the best model. In legends, "+ recipe" denotes our training setup.

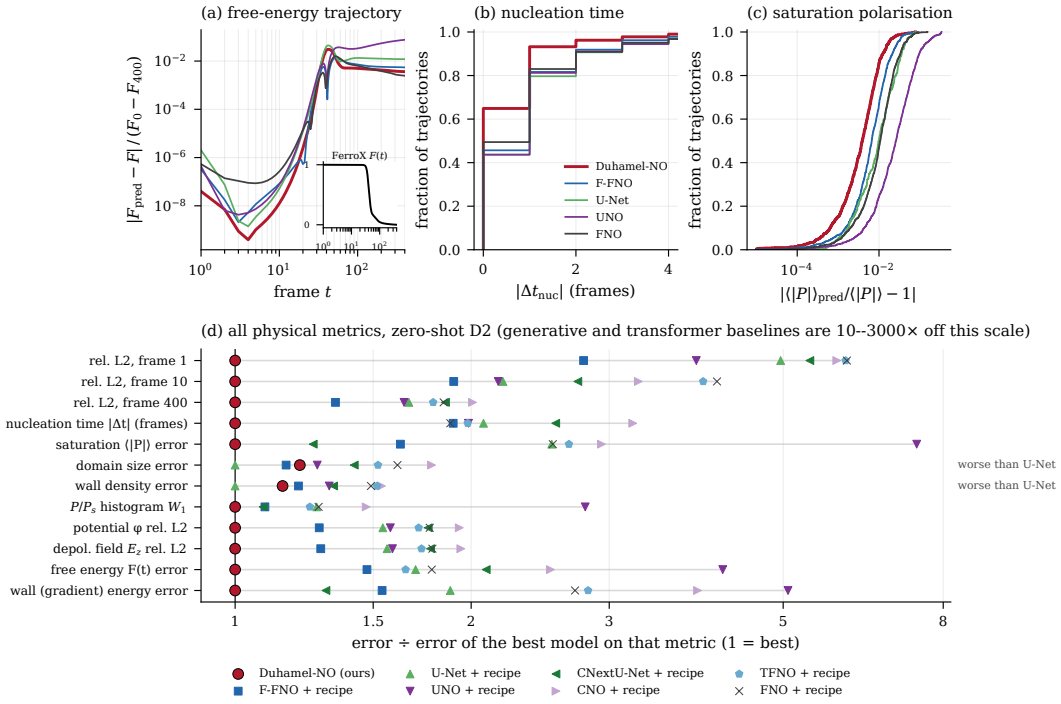


Figure 10: **Physical consistency on the 300 D2 materials** (no retraining, partly outside the training range), as Figure 9.

E.2 Error spectra

Figure 11 shows where in wavenumber each model's error lies, which is the mechanism behind the truncation cap (Proposition 6).

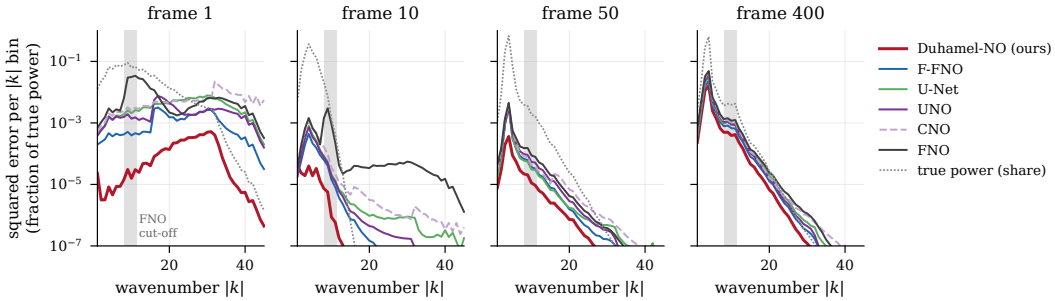


Figure 11: **Error spectra** on D1 held-out materials: squared error per radial wavenumber bin, as a fraction of the total true power, at four frames; grey dotted: the true power spectrum. Shaded: the FNO's mode cut-off. At frame 1 every baseline's error is spread over all wavenumbers; by frame 10 the FNO's error is concentrated above its cut-off.

E.3 Ablations

Table 8 isolates the parts of Duhamel-NO, and Table 9 isolates the change of units.

Table 8: **Ablations of Duhamel-NO** (D1 held-out; mean ± std over seeds), all with the same 9.5M-parameter UNO. Adding the Duhamel step with the measured Fourier SSM gives the gain. A learned, conditioned symbol, which cannot couple the depth layers, is worse than no Duhamel step. A strictly $O(v^2)$ correction keeps frame 1 accurate but underfits late coarsening.

FerroX TDGL (same 9.5M-parameter UNO backbone)	corr@400 (D1)	rel. L2@400 (D1)	rel. L2, frame 1
UNO, no Duhamel step	0.962±0.003	0.219±0.012	0.315±0.005
+ Duhamel step, measured operator (Duhamel-NO)	0.982±0.001	0.129±0.003	0.082±0.000
Duhamel-NO with an $O(u^2)$ remainder instead of UNO	0.934±0.013	0.287±0.042	0.085±0.000
+ Duhamel step, learned conditioned symbol	0.950±0.001	0.256±0.003	0.399±0.026
Duhamel-NO, uniform mode kept instead of damped	0.980	0.133	0.093
Duhamel, learned generator + small $O(u^2)$ network	0.812±0.002	0.578±0.003	0.219±0.001

Table 9: **The change of units** (mean ± std over three seeds): FNO and UNO with our training setup, with and without material units. $\mathrm{B}_{\mathrm{units}}$: D2 materials with P_{S} up to $2.2\times$ the training maximum. The units carry the amplitude out of range ($0.24 \to 0.02$ for FNO, $0.26 \to 0.02$ for UNO); they do not improve the pattern in range.

backbone	change of units	D1 rel. L_2	D1 amplitude err.	D2 $\mathrm{B}_{\mathrm{units}}$ rel. L_2	D2 $\mathrm{B}_{\mathrm{units}}$ amplitude err.
FNO	yes	0.270 ±0.015	0.012 ±0.004	0.207 ±0.006	0.015 ±0.005
FNO	no	0.245 ±0.008	0.018 ±0.000	0.335 ±0.009	0.243 ±0.007
UNO	yes	0.219 ±0.012	0.025 ±0.007	0.163 ±0.013	0.019 ±0.002
UNO	no	0.199 ±0.004	0.049 ±0.002	0.330 ±0.006	0.258 ±0.002

E.4 Cost and training data

Figure 12 relates accuracy to inference cost and to the number of training materials.

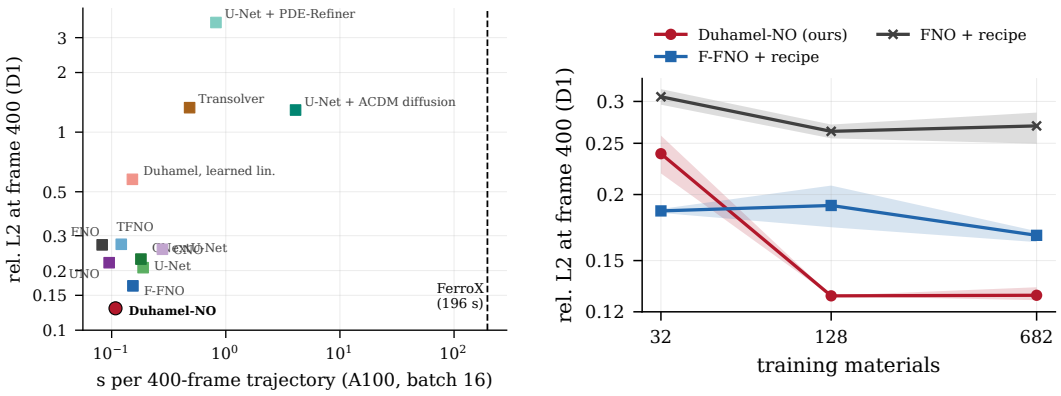


Figure 12: **Left:** error at frame 400 against wall-clock cost per 400-frame trajectory (A100, batch 16), with the FERROX cost. **Right:** error against the number of training materials (the Fourier SSM is rebuilt from the same subset). With 128 materials Duhamel-NO is unchanged (0.129); with 32 it degrades to 0.239, where F-FNO is better (0.186).

E.5 Rollouts of every model

Figures 13 and 14 show one D1 held-out and one D2 trajectory for every model, from the same unseen noise. Each is the trajectory with the median Duhamel-NO error among the saved examples.

F Voltage-driven hysteresis

Setup. The same capacitor, with $\varepsilon_{\mathrm{de}} = 100$ (with 10 the loop is too narrow to resolve), is driven by a quasi-static voltage sweep $0 \to +V_{\max} \to -V_{\max} \to +V_{\max}$ from the unpoled random field. The top electrode potential is stepped in 0.125 V increments, each held until the electrostatic solve converges, with $V_{\max} = 1.0$, 1.5 or 2.0 V and frames every 20 solver steps; a loop takes 440–2765 frames. Every model is retrained on 90 closed loops and rolls out the whole sweep from the first frame, with the voltage as an input. The loop error is the mean absolute deviation of the polarization–voltage curve, in units of P_{S}.

Results. CNO is the best model on held-out materials (0.078; Table 10). Duhamel-NO is second (0.138), with 1.76× the loop error of CNO, although its first-frame error is 4.34× lower than that of the best other model. Its Fourier SSM is measured around the unpoled state, whereas during a sweep the film spends most of its time near saturation, where the linearization is different; and at 20 solver steps per frame only part of the wavenumber bins, not including the uniform mode that carries the switched polarization, can be fitted. Voltage-driven switching is therefore the task on which Duhamel-NO is least competitive. We test on 30 loops of held-out materials, on new noise for training materials, and on an independent set of ten closed loops. Figure 15 shows the deviation of the predicted polarization along the sweep, and Figure 16 the predicted loops of held-out materials.

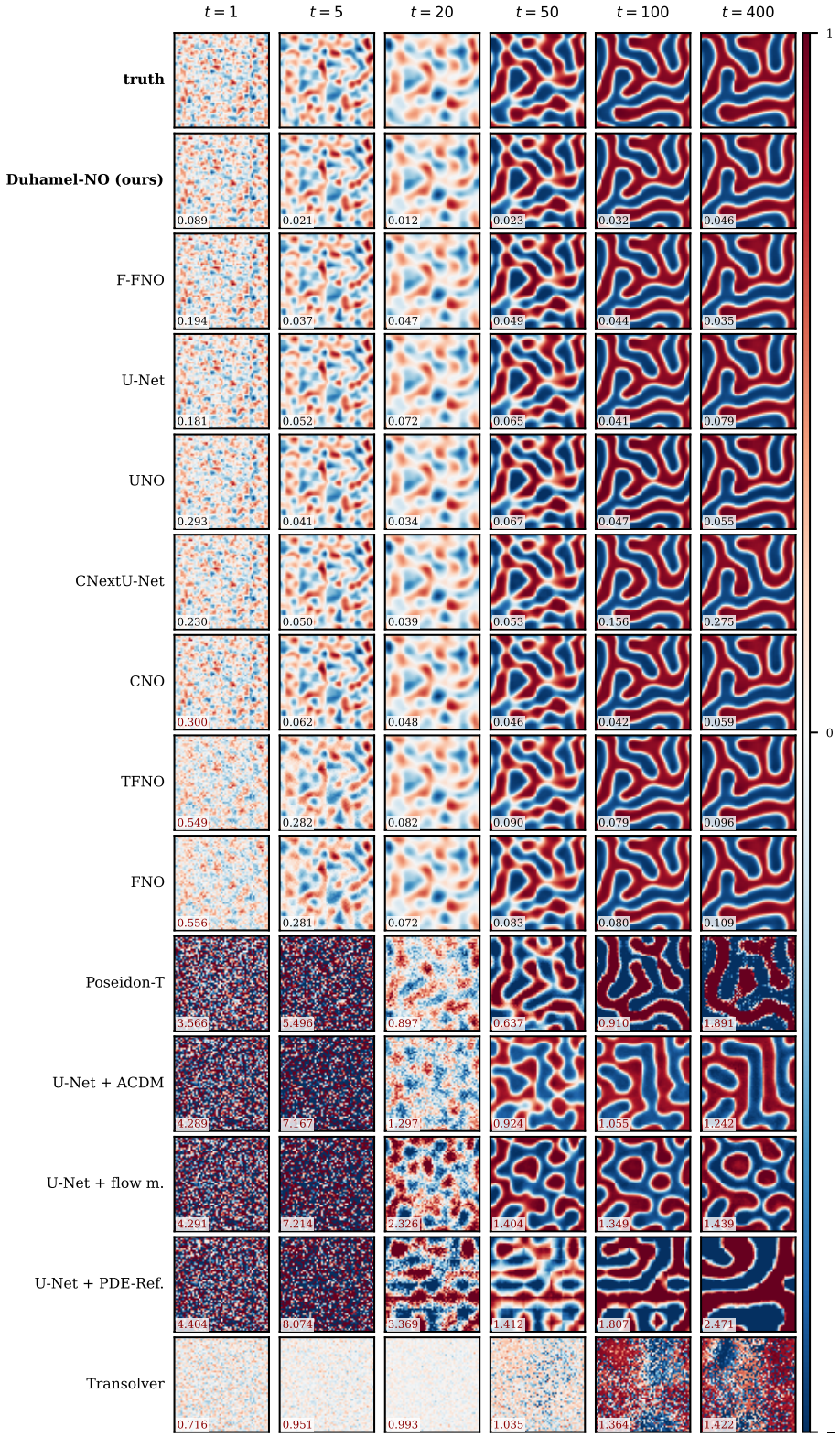


Figure 13: **D1 held-out material, every model** (top ferroelectric layer; field divided by the maximum of the truth at that frame; numbers: relative error of the frame, red above 0.3).

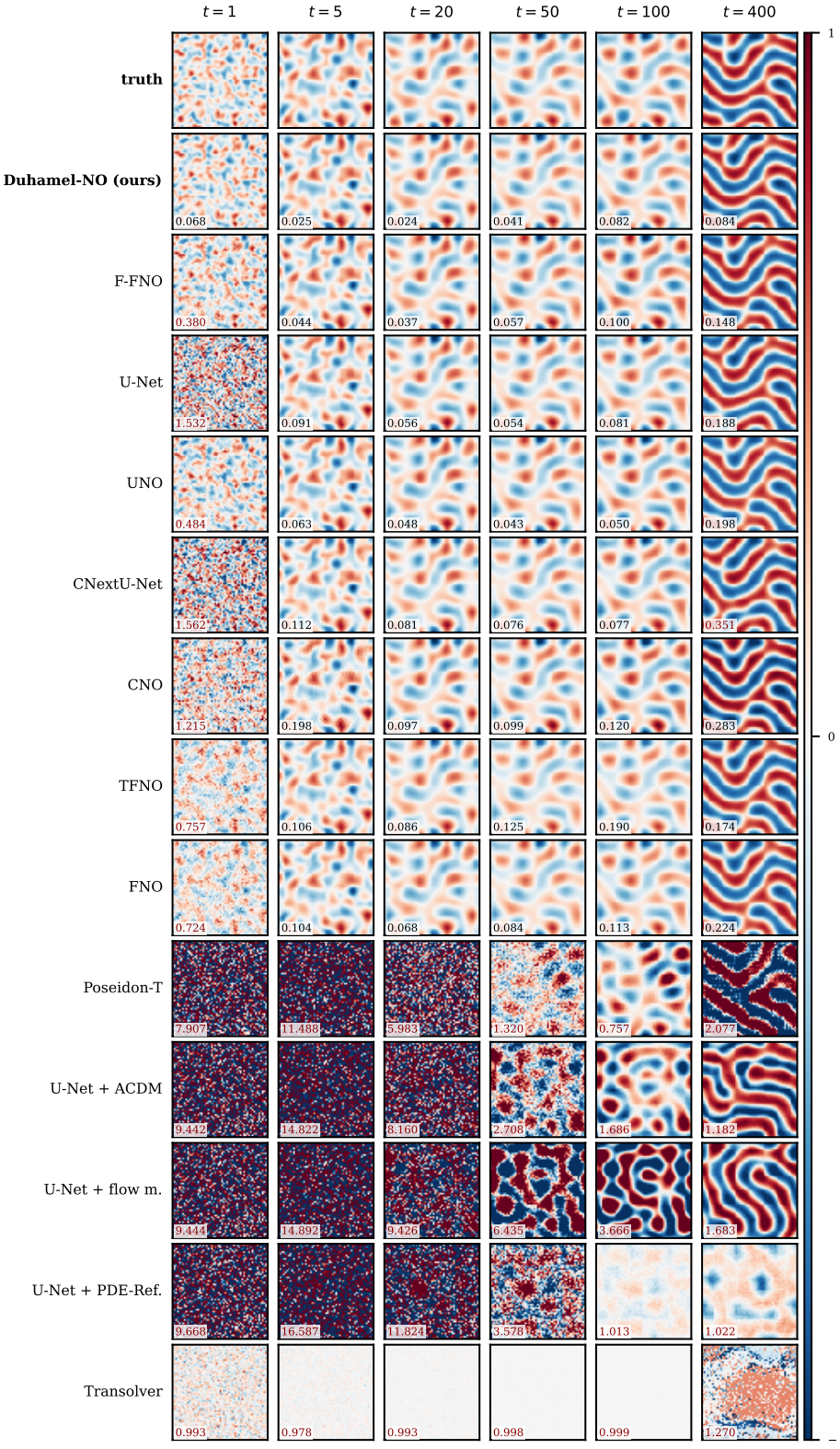


Figure 14: **D2 material (out of range, no retraining), every model**, as Figure 13.

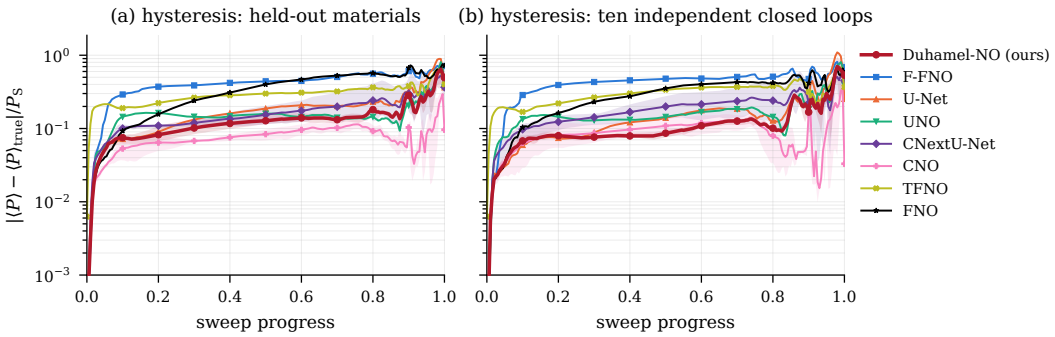


Figure 15: **Hysteresis error along the sweep.** Deviation of the mean polarization from FERROX, in units of P_S, on held-out materials (a) and on ten independent closed loops (b); its time mean is the loop error. Mean over loops; bands: range over training seeds.

Table 10: **Hysteresis, closed loops:** loop error (mean ± std over seeds) on held-out materials, on new noise for training materials, and on an independent set of ten closed loops; field correlation on held-out materials.

model	seeds	held-out material	new noise	extra closed set	field corr. (held-out mat.)
CNO	3	0.078±0.004	0.094±0.012	0.092±0.019	0.722±0.007
Duhamel-NO (ours)	3	0.138±0.014	0.128±0.003	0.121±0.012	0.617±0.042
UNO	1	0.164	0.159	0.172	0.189
CNextU-Net	3	0.171±0.052	0.184±0.055	0.191±0.064	0.455±0.120
U-Net	1	0.194	0.183	0.165	0.174
TFNO	1	0.295	0.310	0.309	0.074
FNO	1	0.361	0.245	0.310	0.397
F-FNO	1	0.431	0.409	0.439	0.264

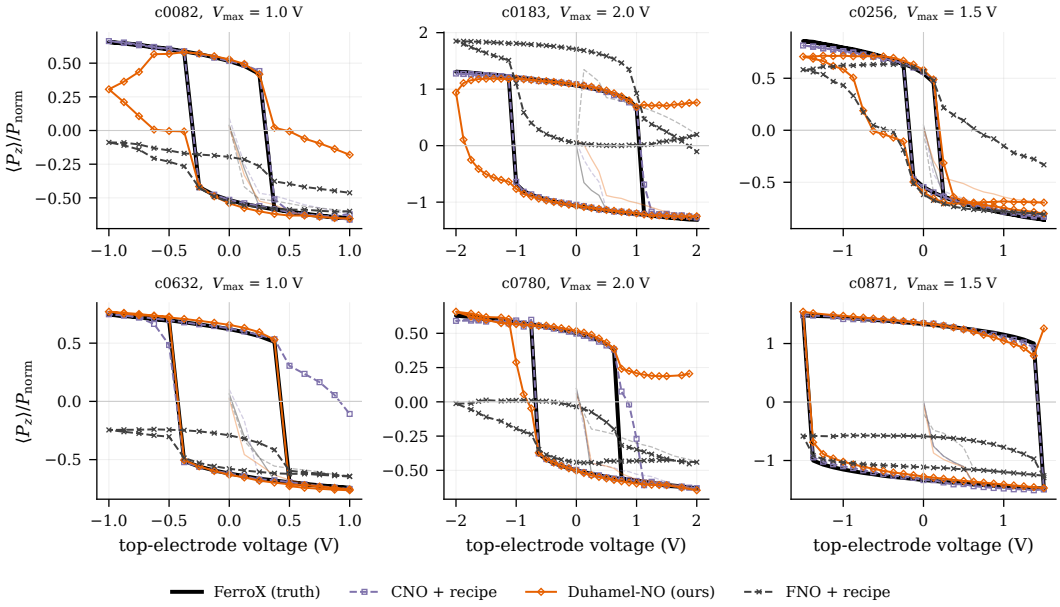


Figure 16: **Closed polarization–voltage loops of held-out materials**, predicted by rollout of the whole sweep from the unpoled random field (virgin branch drawn lightly).

G Allen–Cahn

Setup. Each split has 900 training, 100 validation and 128 test trajectories. The amplitude unit $U = \sqrt{|a|/b}$, the wall width $\mu = \sqrt{e/|a|}$ and the grid spacing in wall widths $\Pi_h = \Delta x/\mu$ are sampled for training ($U \in [0.5, 2]$, $|a| \in [0.5, 5]$, $\Pi_h \in [0.30, 0.70]$). Two test splits leave this range: *new units* ($U \in [4, 10]$, $|a| \in [20, 100]$, same Π_h) and *new physics* ($\Pi_h \in [0.85, 1.20]$). The in-range test materials are also simulated on 32^2 and 128^2 grids of the same domain, and every model trained on 64^2 is rolled out on them unchanged. Table 11 gives the final-frame errors on every split and grid.

Table 11: **Allen–Cahn:** relative L_2 error at the final frame (mean over seeds; **best**, second). New units: amplitude and time units outside the training range. New physics: the wall width relative to the grid outside it. Zero-shot grids: in-range test materials on 32^2 and 128^2, with models trained on 64^2.

	trained grid (64^2)			zero-shot grid (in range)	
model	in range	new units	new physics	32^2	128^2
Duhamel-NO (ours)	**0.012**	**0.010**	**0.075**	0.124	**0.010**
F-FNO	0.043	0.042	0.416	**0.049**	0.086
CNextU-Net	0.057	0.051	1.023	0.879	0.909
U-Net	0.088	0.066	0.378	0.857	0.854
CNO	0.097	0.090	0.667	0.884	0.859
UNO	0.092	0.090	0.920	0.717	0.194
TFNO	0.120	0.102	0.265	0.094	0.271
FNO	0.150	0.148	0.443	0.115	0.276